```
  GNU nano 7.2
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long counter = 0;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++)
        counter++;
    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Expected counter: %d\n", THREADS * ITERATIONS);
    printf("Actual counter: %ld\n", counter);

    return 0;
}
```

```
  GNU nano 7.2
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long counter = 0;
pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }
    return NULL;
}

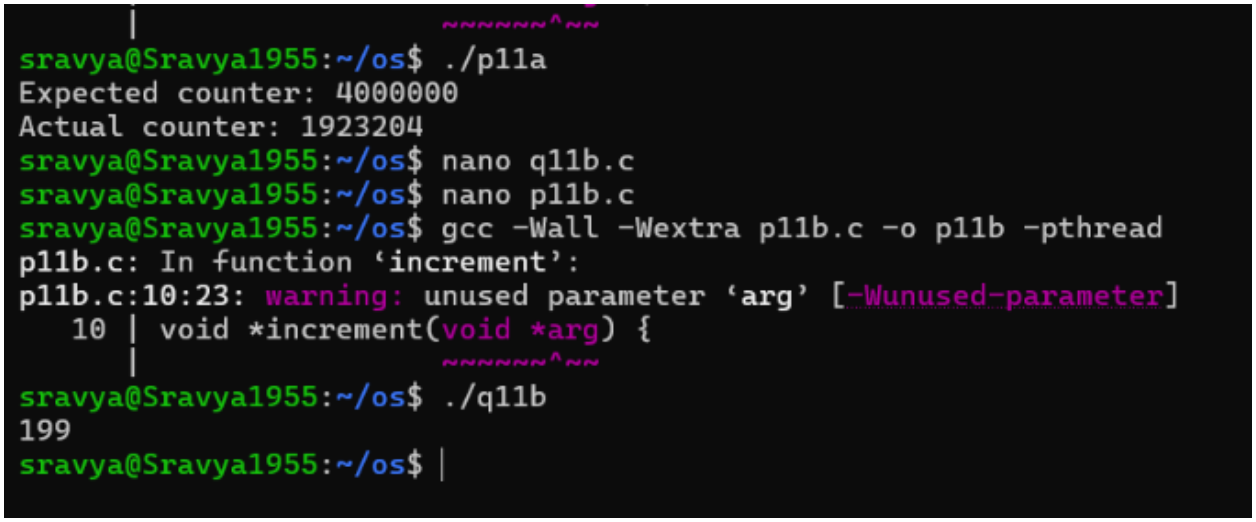
int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Expected counter: %d\n", THREADS * ITERATIONS);
    printf("Actual counter: %ld\n", counter);

    pthread_mutex_destroy(&lock);
    return 0;
}
```


~~~~~~^~~
sravya@Sravya1955:~/os$ ./p11a
Expected counter: 4000000
Actual counter: 1923204
sravya@Sravya1955:~/os$ nano q11b.c
sravya@Sravya1955:~/os$ nano p11b.c
sravya@Sravya1955:~/os$ gcc -Wall -Wextra p11b.c -o p11b -pthread
p11b.c: In function ‘increment’:
p11b.c:10:23: warning: unused parameter ‘arg’ [-Wunused-parameter]
10 | void *increment(void *arg) {
~~~~~~^~~
sravya@Sravya1955:~/os$ ./q11b
199
sravya@Sravya1955:~/os$